Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Newton’s second law

Meaning in my words What does Newton’s second law tell us about a cart?	**How can you tell?** Why add the forces before dividing by mass?
Example or sketch A 2 kg cart feels 7 N right and 3 N left. What acceleration do you predict?	**Non-example and why** A partner uses 7 N alone to find acceleration. Which part of the diagram contradicts that?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Direction

Meaning in my words	How can you tell?
How would you explain direction to a friend?	A cart moves right while its net force points left. Which way does acceleration point?
Example or sketch	**Non-example and why**
Draw arrows for a cart moving left and slowing down.	"The biggest force points east, so the cart must be moving east." What can we actually tell if the net force is east?

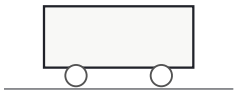
Balanced

Meaning in my words	How can you tell?
What does it mean to say the forces on a cart are balanced?	A cart feels 5 N left and 5 N right. How can you tell its horizontal forces balance?
Example or sketch	**Non-example and why**
A book rests on a table. What two vertical arrows would you draw?	One cart pushes another, and the second pushes back. Do these two forces balance on the first cart?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.



My marks mean

On assignment: Net force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A tracking model moves left. Its 4 kg body feels 12 N right and 4 N left. The status display has separate motion and force arrows.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Controls engineer**

Set the net-force and acceleration arrows. Tell the operator what happens to speed at first while velocity points left.

☐ **Replay analyst**

Annotate the slow-motion clip with one arrow for motion and another for acceleration. Explain why they can point opposite ways.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.